

RAJNISH KUMAR

Jamui, Bihar, 811312

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Summary

Software Developer specializing in Generative AI and LLM-powered applications. Built a multi-agent stock analysis platform (70% faster insights), an AI librarian with 750ms semantic search across 16K+ books for 10k+ monthly users, and AgroVision — a microservices-based agricultural platform with multilingual RAG chatbot and ML-powered crop & disease detection. Proficient in React, Node.js, FastAPI, LangChain, LangGraph, and FAISS. 500+ DSA problems solved, 5-Star HackerRank rating.

Technical Skills

Languages: Python, C++, JavaScript

AI & LLM: GenAI, LangChain, LangGraph, RAG, FAISS, Gemini AI, Whisper, Hugging Face, Sentence Transformers

ML & CV: Numpy, Pandas, PyTorch, TensorFlow, Scikit-Learn, OpenCV, YOLO Computer Vision

Backend: FastAPI, Node.js, Express.js, REST APIs

Frontend: React, Tailwind CSS

Databases: PostgreSQL, MongoDB, Redis, ChromaDB, FAISS

DevOps & Tools: Docker, JWT, PASETO, RazorPay, SMTP

Projects

AI Multi-Agent Stock Analysis (Finlyze) | [Live](#) | *LangGraph, Gemini AI, Streamlit* Jan 2026 – Feb 2026

- Masterminded a **LangGraph-driven** multi-agent ecosystem, coordinating specialized Researcher, Analyst, and Writer agents to quantify technicals and sentiment, accelerating insight discovery by 70%.
- Automated a high-volume data pipeline integrating **yfinance** and **DuckDuckGo** to aggregate 50+ technical indicators and 150+ daily news reports for **Gemini-powered** NLP sentiment analysis.
- Modernized a **Streamlit**-based analytics hub on Render, utilizing asynchronous concurrency and **fpdf2** for automated PDF report generation, achieving consistent sub-2s reporting speeds.

LitScholar AI-Powered Virtual Librarian | *FastAPI, React, FAISS, Neon PostgreSQL* Feb 2026

- Built **AI librarian** with semantic search over 16k+ books, enabling natural language queries and follow-ups with 65% lower latency and 92% citation accuracy.
- Developed **recommendation** engine using **collaborative filtering** and **genre similarity**, achieving 95% preference prediction accuracy.
- Implemented RAG pipeline with 750ms retrieval using **Sentence Transformers**, FAISS, and React serving 10k+ monthly users.
- Designed **microservices** architecture with four backends, integrating **Razorpay** subscriptions and **Google OAuth** authentication.

AgroVision - AI Agricultural Platform | *React, FastAPI, Node.js, TensorFlow, LangChain, FAISS, MongoDB* 2025 – 2026

- Unified **crop recommendation**, **disease detection**, and voice assistance into a **PASETO**-secured microservices platform with a responsive **multilingual** dashboard (English, Hindi, Punjabi) built for low-literacy farming users using **React**.
- Engineered **KrishiBot**—a **RAG**-powered chatbot (**Gemini + FAISS**) with OpenAI **Whisper STT** and **gTTS**, limiting LLM context tokens via **retrieval** and supporting text and voice queries in English, Hindi, and Punjabi.
- Deployed **EfficientNetB0 transfer learning** model (**ImageNet** backbone, 38-class custom head) achieving **96.5%** accuracy, improved to **97.1%** after **fine-tuning**, alongside a scikit-learn crop recommendation pipeline using live **Open-Meteo** weather data.

Certifications

Complete Machine Learning and Data Science (GeeksForGeeks)

Introduction to Large Language Models (Coursera)

Complete Computer Vision Bootcamp with PyTorch & TensorFlow (Udemy)

Achievements

Solved 500+ algorithmic problems across LeetCode, GeeksforGeeks, Codeforces, and HackerRank.

Achieved a peak LeetCode rating of 1364 through consistent participation in global competitive programming contests.

Ranked 5-Star on HackerRank, validating advanced proficiency in Python syntax and algorithmic efficiency.

Education

Lovely Professional University

Oct 2022 – Present

Bachelor of Technology in Computer Science and Engineering; CGPA: 7.48

Phagwara, Punjab

Red Rose School

Mar 2019 – Jun 2021

Senior Secondary (Science); Percentage: 84%

Deoghar, Jharkhand